

Use-Case Flow Specification

Use Case ID: UC-04

Use Case Name: Generate Audit Export

Primary Actors:

- Compliance Officer
- Security Auditor

Goal:

Generate an immutable and cryptographically verifiable export containing selected audit records and their associated integrity information.

Preconditions

1. The actor has access to the Centralized Audit Trail Compliance Engine.
2. Audit records are available in the system.
3. The actor has selected one or more audit records for export.

Trigger

The actor selects the **Generate Audit Export** option for the chosen audit records.

Main Success Scenario

1. The actor selects the audit records that need to be exported.
2. The actor requests the system to generate an audit export.
3. The system validates that the selected audit records are available.
4. The system retrieves the selected audit records from storage.
5. The system obtains the integrity information associated with the selected records.
6. The system creates an audit export containing the selected records and their integrity information.
7. The system finalizes the export so that its integrity can later be cryptographically verified.
8. The system displays a confirmation that the audit export was generated successfully.
9. The generated audit export is made available to the actor.

Postconditions

1. An audit export containing the selected records has been generated.
2. Integrity information required to verify the export is included.
3. The original stored audit records remain unchanged.
4. Any later modification of the export can be detected through integrity verification.

Alternate Flow — No Valid Records Selected

3a. The system determines that no valid audit records are available in the current selection.

1. The system does not generate an audit export.
2. The system displays a message indicating that no valid records were selected.
3. The actor may return to **Search & Filter Audit Logs (UC-03)** and select appropriate audit records.
4. The use case ends without generating an export.